SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME:	Cloud Computing and Big Data Analytics	**COURSE CODE:**	CSA15

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 50

Code of Conduct

I am Gurusurya.G (192311332) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Gurusurya

Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded:

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.

ANSWER:

Feature	Type-1 Hypervisor	Type-2 Hypervisor
Installation	Runs directly on hardware	Runs on top of a host operating system
Performance	Very high	Lower due to OS overhead
Security	More secure	Less secure because host OS can be attacked
Resource Usage	Efficient	Uses extra system resources
Examples	VMware ESXi, Microsoft Hyper-V, Xen	VMware Workstation, Oracle VirtualBox

Analysis:

- **Performance:** Type-1 provides faster execution because it directly controls hardware.
- **Security:** Type-1 has fewer attack points since there is no host operating system.
- **Manageability:** Type-1 supports centralized management, live migration, and high availability.

Justification:

For a mission-critical cloud datacenter, Type-1 Hypervisor is the better choice because it offers:

- Better performance
- Higher security
- Better scalability
- Improved reliability

2. **A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.**

ANSWER:

A cloud provider hosts finance, healthcare, and student applications on the same hardware.
Isolation Support
Virtualization creates separate Virtual Machines (VMs) where each workload has:

- Independent operating system
- Dedicated virtual CPU, RAM, and storage
- Separate execution environment

This prevents one application from affecting another.

Two Major Attack Surfaces:

1. **Hypervisor Attack:**

If the hypervisor is compromised, all VMs may be affected.
Solution

- Secure hypervisor
- Regular updates
- Access control

2. **VM Escape Attack:**

A malicious VM attempts to access another VM or the host system.

Solution:

- Strong VM isolation
- Network segmentation
- Security monitoring

3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.

ANSWER:

Given

- CPU = 92%
- Memory = 88%
- Storage I/O latency = 35 ms
- VM boot time = 170 s

Likely Bottlenecks:

CPU Bottleneck:
CPU utilization above 90% causes slower application response.

Memory Bottleneck:
High memory usage may lead to swapping, reducing performance.

Storage Bottleneck:
35 ms latency indicates slow disk performance.

Boot Delay:
170 seconds indicates overloaded storage or insufficient resources.

Corrective Actions

- Add more CPU cores
- Increase RAM
- Upgrade to SSD/NVMe storage
- Balance workloads across additional hosts
- Enable resource monitoring and auto-scaling

4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.

ANSWER:

Traditional Datacenter	**Software Defined Datacenter**
Physical hardware management	Software-controlled resources
Manual configuration	Automated provisioning
Limited scalability	Highly scalable
Slower deployment	Faster deployment

Compute Virtualization:

- Creates multiple virtual machines on one server.
- Improves hardware utilization.

Storage Virtualization:

- Combines multiple storage devices into a single storage pool.
- Enables faster backup and flexible allocation.

Network Virtualization:

- Creates virtual networks independent of physical hardware.
- Improves security using virtual firewalls and network isolation.

Conclusion:

SDDC provides:

- Better agility
- Higher scalability
- Improved resource utilization
- Lower operational cost

5.A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

ANSWER:

Users cannot log in across multiple cloud providers because their identities do not match.

Causes:

- Different authentication systems
- Separate user databases
- Inconsistent identity management.

Secure Identity and Privacy Design

Identity Federation:
Use a common Identity Provider (IdP) to authenticate users once.

Single Sign-On (SSO):
Users log in once and access multiple cloud services.

Secure Authentication:
Use:

- Multi-Factor Authentication (MFA)
- OAuth 2.0
- SAML
- OpenID Connect (OIDC)

Privacy Protection:

- Encrypt user data (AES-256)
- Apply role-based access control (RBAC)
- Share only necessary user information

Conclusion:

Federated identity enables:

- Secure authentication
- Better privacy
- Easier access across multiple cloud providers.